Semester project on control and simulation of automation systems

BEng in Robot Systems
4. Semester

Authors (Group 1?)

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Total characters:

May 12, 2026

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Contents

1 Introduction

In life and industri that are alot of machinery that need regualted and precice controll, which i why a controller is made to achive each goal as needed. In robotics controllers can be expicaly usefull when trying to complete difficult taskes with multipul moveing parts. Where balance is one of main hindrences, when trying to design such a system. This projeckt will focus trying to solve a balance isuee of an inverted pendulem.

1.1 Project Goals

With a PLC, motor, conveyor track, inverted pendulem on a cart and a controller. it is attempted to make the inverted pendulem balance rightsideup using a controller to controll the force of the motor. success is mesurd by how well the controller balences the pendulem compared to a simulation of the controller.

1.2 Problem Definition

1.3 Constraints & Limitations

The limitations for this projeckt come from the physical setup, and the PLC. The cart is limited to move within two sensors on the conveyor track, and the maximum force for the moter is X. For the PLC it cant run faster than 1000Hz, and sampling via OPCUA is limited to 100Hz

1.4 System OverView

2 System modeling

3 Control theroy

4 PLC implemetation

5 Data Collection

6 Evaluation

7 Discussion

8 Conclusion

References

A Distribution of Tasks

Table A.1: Distribution of Tasks

Task	**Noah**	**Rasmus**	**Emma**	**Jannick**	**Eymundur**
Development					
System Modeling					
Control Theroy					
PLC implementation					
Data Collection					
Report					
Abstract					
Introduction					
System Modeling					
Control theroy					
PLC Implementation					
Data Collection					
Results					
Discussion					
Conclusion					

B Code

The code is available in the attached ZIP file as well as on github at:

C Video Showcase

A video of the system running is available at: